Akshay Shivashankar Bharadwaj

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Education

University of Southern California, MS in Computer Science (GPA 3.8) Jan 2025 – Dec 2026

- **Courses:** Analysis and Design of Algorithms, Web Technologies, Operating Systems, Cloud and Distributed Systems

BMS College of Engineering, BE in Computer Science (GPA 4.0) Aug 2018 – Jun 2022

- **Courses:** Data Structures and Algorithms, Web and Database design, Artificial Intelligence, Machine Learning

Experience

IT Engineer, Qualcomm - Bengaluru, India June 2022 – Dec 2024

- Engineered an **AngularJS–Flask** internal tool integrating **ServiceNow APIs** to automate end-to-end RFC (request for change) workflows, eliminating **100+ engineering hours of manual effort per month**.
- Built end-to-end containerized microservices infrastructure using **Docker, Kubernetes (EKS)**, and AWS CloudFormation (S3, RDS, Route53, VPC, Lambda), cutting deployment time by **50%** and eliminating environment-parity issues.
- Drove **cross-functional collaboration** with Dev, QA, and Infra teams to implement **CI/CD pipelines** via CloudFormation, replacing manual deployments and reducing environment setup errors to **near-zero**.
- Designed and deployed an **AWS CloudFormation automation** framework to onboard **70+ web applications** onto Amazon Linux EC2, incorporating MFA and 24/7 CloudWatch/DataDog alerting to enforce security and availability SLAs.

Interim Intern, Qualcomm - Bengaluru, India Feb 2022 – Jun 2022

- Built a **self-service AWS deployment infrastructure using CloudFormation and Lambda** that enabled teams to deploy Java applications with minimal manual input, cutting new environment setup from days to **under an hour**.

Projects

Operating Systems Kernel Development — Weenix (Unix-Like OS Kernel)

- **Designed and implemented core kernel subsystems** in C, including process and thread management (scheduling, context switching, synchronization), virtual file system (VFS), and system calls within a Linux-inspired OS kernel.
- **Developed virtual memory and process creation mechanisms**, implementing virtual memory mapping, page fault handling, and `fork()`-based child process creation to support concurrent user processes and robust memory isolation.

Raft Consensus Protocol — Distributed Systems (C++20, gRPC)

- Built a production-style implementation of the **Raft consensus leader election and log replication protocols** using gRPC, ensuring safety and liveness under node crashes, network partitions, and delayed RPC delivery.
- Designed concurrent, thread-safe state management for distributed RPC handlers with term quorum-based voting and replicated log consistency, validated through automated fault-injection tests simulating real-world failure scenarios.

Artsy - Mobile and Web Front End using Android Studio and AngularJS (MEAN Stack)

- As part of the Web Technologies course (CSCI-571) I developed a Web and Mobile Front-end deployed on the **Google Cloud Platform (GCP)** for an application mirroring the Artsy application showcasing artist information.
- Utilized **REST APIs, AngularJS** and **Android Studio** with a **NodeJS (Express)** and **MongoDB** backend.

Eye-Move: Eye-Tracking Based Typing Application (Computer Vision)

- Leveraged Python and its Computer Vision libraries, mainly **OpenCV and DLib** for eye blink detection and creation of a virtual keyboard to help motor-neuron disability patients with communication.

Scientific Collaborator - Moonquake Alert System (MAS), Space@yourService - EPFL

- Built a **Python-based** data science application for moonquake detection leveraging seismic data recorded by lunar rovers.
- Coordinated with EPFL university for testing and deploying application on **Docker containers** for evacuation procedures as part of the experiment 'Asclepios' regarding establishing a base on the moon.

Technologies

Languages: Python, C++, C, Java, SQL, JavaScript, PHP, HTML, CSS, Unix Shell Scripting

Core Technologies: AWS **(AWS Certified Solutions Architect - Associate)**, Git, Docker, Kubernetes, Tomcat, Flask, REST APIs, AngularJS, Bootstrap, MongoDB, MySQL, Datadog, PowerBI, Linux Kernel Programming